TauTimp — Entropy Virtual Particle Wave Equation Onduscular Theory (Teoria Onduscular) — Viorel Popescu

1. Master Equation and Setup

The entropy virtual-particle wave equation in TauTimp is:

$$\frac{R}{r}\,\Delta S(r,\theta,t) = \frac{1}{V^2}\frac{\partial^2 S}{\partial t^2}$$

with propagation velocity V depending on the virtual-particle species:

Virtual Particle	Velocity V
Photon (virtual)	$c = 2.998 \times 10^8$ m/s
Electron (virtual)	$v_e \approx c/137$ (Bohr)
Phonon (acoustic, virtual)	$v_s = 346$ m/s
Gamma / transverse wave	$V_\gamma = 10^9\, c$

We solve the prototype equation with $V = c$: $\boxed{\frac{R}{r}\,\Delta f(r,t) = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}}$

Cosmological parameter: $R = 4.2355 \times 10^{22}$ m (onduscular radius).

2. Laplacian in Spherical Symmetry

For $f = f(r,t)$ (no θ, ϕ dependence):

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}$$

The master PDE becomes:

$$\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}\right) = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}$$

3. Separation of Variables

Set $f(r,t) = g(r)\,u(t)$. Substituting:

$$\frac{R}{r}\left(g''u + \frac{2}{r}g'u\right) = \frac{g\,\ddot{u}}{c^2}$$

Dividing both sides by $g\,u$:

$$\frac{R}{r}\cdot\frac{g'' + \frac{2}{r}g'}{g} = \frac{\ddot{u}}{c^2\,u} = -\omega^2 \quad \text{(separation constant)}$$

3.1 Temporal ODE

$$\ddot{u} + \omega^2 c^2\,u = 0 \Longrightarrow u(t) = A\cos(\omega c\,t) + B\sin(\omega c\,t)$$

3.2 Radial ODE

$$g'' + \frac{2}{r}g' + \frac{\omega^2 r}{R}\,g = 0$$

Substitute $g(r) = \psi(r)/r$, so $g' = \psi'/r - \psi/r^2$, $g'' = \psi''/r - 2\psi'/r^2 + 2\psi/r^3$:

$$\frac{\psi''}{r} - \frac{2\psi'}{r^2} + \frac{2\psi}{r^3} + \frac{2}{r}\left(\frac{\psi'}{r} - \frac{\psi}{r^2}\right) + \frac{\omega^2 r}{R}\cdot\frac{\psi}{r} = 0$$

$$\frac{\psi''}{r} + \frac{\omega^2}{R}\,\psi = 0 \Longrightarrow \psi'' + \frac{\omega^2}{R}\,\psi = 0$$

This is **simple harmonic** in r:

$$\boxed{\psi(r) = C_1\cos\left(\frac{\omega r}{\sqrt{R}}\right) + C_2\sin\left(\frac{\omega r}{\sqrt{R}}\right)}$$

Hence:

$$g(r) = \frac{\psi(r)}{r} = \frac{C_1\cos(kr) + C_2\sin(kr)}{r}, \qquad k = \frac{\omega}{\sqrt{R}}$$

4. Boundary Condition and Eigenvalue Quantization

Condition: $f(2R,0) = 0$ with $u(0) = 1$ (absorbed into A) $\Rightarrow g(2R) = 0$.

Regularity at $r \to 0$ forces $C_1 = 0$ (cosine term diverges as $1/r$). Thus:

$$g(r) = \frac{C_2\sin(kr)}{r}$$

Boundary condition at $r = 2R$:

$$g(2R) = \frac{C_2\sin(2kR)}{2R} = 0 \Longrightarrow \sin(2k_nR) = 0 \Longrightarrow 2k_nR = n\pi, \quad n = 1,2,3,\ldots$$

$$\boxed{k_n = \frac{n\pi}{2R}, \qquad \omega_n = k_n\sqrt{R} = \frac{n\pi}{2\sqrt{R}}}$$

Numerical values ($R = 4.2355 \times 10^{22}$ m, $\sqrt{R} = 6.5080 \times 10^{11}$ m$^{1/2}$):

$$k_1 = \frac{\pi}{2R} = \frac{\pi}{8.4710 \times 10^{22}} \approx 3.709 \times 10^{-23}\ \text{m}^{-1}$$

$$\omega_1 = \frac{\pi}{2\sqrt{R}} \approx \frac{\pi}{1.3016 \times 10^{12}} \approx 2.414 \times 10^{-12} \text{ rad/s}$$

5. Full Solution $f(r,t)$

The general solution is a superposition:

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} \frac{\sin(k_n r)}{r} [A_n \cos(\omega_n c\, t) + B_n \sin(\omega_n c\, t)]}$$

with $k_n = \frac{n\pi}{2R}$ and $\omega_n = \frac{n\pi}{2\sqrt{R}}$.

For the fundamental mode ($n = 1$): $f_1(r,t) = \frac{\sin(k_1 r)}{r} [A_1 \cos(\omega_1 c\, t) + B_1 \sin(\omega_1 c\, t)]$

Period of fundamental oscillation: $T_1 = \frac{2\pi}{\omega_1 c} = \frac{4\sqrt{R}}{c} \approx \frac{4 \times 6.508 \times 10^{11}}{2.998 \times 10^8} \approx 8.683 \times 10^3 \text{ s} \approx 2.41 \; hours$

6. Angular Extension: Building $S(r,\theta,t)$

For full spherical symmetry with angular dependence, expand in Legendre polynomials:

$$S(r,\theta,t) = \sum_{n=1}^{\infty} \sum_{\ell=0}^{\infty} \frac{\sin(k_n r)}{r} P_\ell(\cos\theta) [A_{n\ell} \cos(\Omega_{n\ell} t) + B_{n\ell} \sin(\Omega_{n\ell} t)]$$

where the effective frequency includes the angular eigenvalue $\ell(\ell+1)$:

$$\Omega_{n\ell} = c \sqrt{\omega_n^2 + \frac{\ell(\ell+1)}{r_0^2}}$$

For the spherically symmetric case ($\ell = 0, P_0 = 1$):

$$S(r,\theta,t) = f(r,t) = \sum_{n=1}^{\infty} \frac{\sin(k_n r)}{r} [A_n \cos(\omega_n c\, t) + B_n \sin(\omega_n c\, t)]$$

7. Probabilistic Phonon Flux — Density $P(r,t)$ 7.1 Governing Equation

The phonon probability density flux satisfies the d'Alembert equation:

$$\frac{1}{c^2} \frac{\partial^2 P}{\partial t^2} - \frac{\partial^2 P}{\partial r^2} = 0$$

with flux density $\frac{P(r,t)}{4\pi r^2}$.

Boundary condition: $P(2R, 0) = 0$.

Additional regularity: $\frac{\partial P}{\partial r}|_{r=0} = 0$ (no flux gradient at origin).

7.2 Gaussian Solution with Polynomial-Quadratic Profile (at $t = 0$)

We seek $P(r,t)$ quadratic in r with Gaussian envelope at $t = 0$.

Ansatz at $t = 0$:

$$P(r,0) = \mathcal{N}\, (a_0 + a_1 r + a_2 r^2) \exp\left(-\frac{r^2}{2\sigma^2}\right)$$

Applying $\partial P / \partial r|_{r=0} = 0$: coefficient of linear term at origin must vanish, so $a_1 = 0$.

Applying $P(2R, 0) = 0$: $(a_0 + a_2 (2R)^2) \exp\left(-\frac{4R^2}{2\sigma^2}\right) = 0 \Rightarrow a_0 = -4a_2 R^2$

Setting $a_2 = 1$ (normalization absorbed into $\mathcal{N}$): $\boxed{P(r,0) = \mathcal{N}\, (r^2 - 4R^2) \exp\left(-\frac{r^2}{2\sigma^2}\right)}$

(Valid for $0 \le r \le 2R$; negative sign ensures $P \ge 0$ in this domain when $\mathcal{N} < 0$, or we take $|P|$.)

Taking $\sigma = R$ as the natural scale:

$$P(r,0) = \mathcal{N}\, (r^2 - 4R^2)\, e^{-r^2/(2R^2)}$$

7.3 Full Space-Time Solution via d'Alembert

The general solution of $P_{tt}/c^2 - P_{rr} = 0$ is:

$$P(r,t) = F(r - ct) + G(r + ct)$$

From $P(r,0) = h(r)$ and $\partial_t P(r,0) = 0$ (standing wave):

$$P(r,t) = \frac{1}{2} [h(r - ct) + h(r + ct)]$$

where $h(r) = \mathcal{N}(r^2 - 4R^2)\, e^{-r^2/(2R^2)}$.

Explicit form: $\boxed{P(r,t) = \frac{\mathcal{N}}{2} \left[((r-ct)^2 - 4R^2) e^{-(r-ct)^2/(2R^2)} + ((r+ct)^2 - 4R^2) e^{-(r+ct)^2/(2R^2)} \right]}$

7.4 Normalization

The normalization constant $\mathcal{N}$ is fixed by:

$$\int_0^{2R} \frac{P(r,0)}{4\pi r^2} \cdot 4\pi r^2\, dr = 1 \Rightarrow \mathcal{N} \int_0^{2R} (r^2 - 4R^2)\, e^{-r^2/(2R^2)}\, dr = 1$$

The integral (with $\sigma = R$):

$$I = \int_0^{2R} (r^2 - 4R^2) e^{-r^2/(2R^2)}\, dr$$

Let $u = r/(\sqrt{2}R)$: $r^2 = 2R^2u^2$, $dr = \sqrt{2}R\, du$, upper limit $u = \sqrt{2}$:

$$I = \sqrt{2}R \int_0^{\sqrt{2}} (2R^2u^2 - 4R^2) e^{-u^2}\, du = 2\sqrt{2}R^3 \int_0^{\sqrt{2}} (u^2 - 2) e^{-u^2}\, du$$

Using standard Gaussian integrals:

$$\int_0^{\sqrt{2}} e^{-u^2}\, du = \frac{\sqrt{\pi}}{2} \operatorname{erf}(\sqrt{2}) \approx \frac{\sqrt{\pi}}{2}(0.9953) \approx 0.8821$$

$$\int_0^{\sqrt{2}} u^2 e^{-u^2}\, du = \frac{\sqrt{\pi}}{4} \operatorname{erf}(\sqrt{2}) - \frac{\sqrt{2}}{2} e^{-2} \approx 0.4411 - 0.09561 \approx 0.3455$$

$$I = 2\sqrt{2}R^3(0.3455 - 2 \times 0.8821) = 2\sqrt{2}R^3(0.3455 - 1.7642) = 2\sqrt{2}R^3(-1.4187) \approx -4.014\, R^3$$

$$\mathcal{N} = \frac{1}{I} \approx \frac{-1}{4.014\, R^3} \approx -\frac{0.2491}{R^3}$$

With $R = 4.2355 \times 10^{22}$ m, $R^3 \approx 7.595 \times 10^{67}$ m^3:

$$\mathcal{N} \approx -3.280 \times 10^{-69}\ \text{m}^{-3}$$

8. Physical Interpretation — Virtual Particle Flux

The phonon flux density at radius r and time t:

$$\mathcal{F}(r,t) = \frac{P(r,t)}{4\pi r^2}$$

At $t = 0$ (snapshot):

$$\mathcal{F}(r,0) = \frac{\mathcal{N}(r^2 - 4R^2)\, e^{-r^2/(2R^2)}}{4\pi r^2}$$

The virtual particle interpretation: the entropy quantum is transmitted as a **"hole"** (absence of occupation in the field lattice) — analogous to a hole in a semiconductor. The propagation velocity determines the species:

Species	V	Nature
Virtual photon	c	Electromagnetic void
Virtual electron	$c/137$	Electronic vacancy
Virtual phonon	346 m/s	Acoustic vacancy
Virtual gamma	$10^9 c$	Transverse field quantum

9. Summary Table

Quantity	Formula	Numerical Value
Cosmological radius	R	4.2355×10^{22} m
Eigenwavenumber	$k_n = n\pi/(2R)$	$k_1 = 3.709 \times 10^{-23}$ m^{-1}
Angular frequency	$\omega_n = n\pi/(2\sqrt{R})$	$\omega_1 = 2.414 \times 10^{-12}$ rad/s
Fundamental period	$T_1 = 4\sqrt{R}/c$	8.683×10^{3} s
Radial solution	$g_n(r) = \sin(k_n r)/r$	sinc-type
Gaussian width	$\sigma = R$	4.2355×10^{22} m
Normalization $\mathcal{N}$	$-1/(4.014R^3)$	-3.280×10^{-69} m^{-3}

10. Connection to Entropy Field $S(r,\theta,t)$

The full entropy field is assembled as:

$$S(r,\theta,t) = f(r,t) \cdot \Xi(\theta)$$

where for spherical symmetry $\Xi(\theta) = 1$ ($\ell = 0$ mode). The entropy flux at radius r is:

$$\frac{S(r,\theta,t)}{4\pi r^2} = \frac{f(r,t)}{4\pi r^2} = \frac{1}{4\pi r} \sum_{n=1}^{\infty} \sin(k_n r)[A_n \cos(\omega_n ct) + B_n \sin(\omega_n ct)]$$

The probability density of finding an entropy quantum in $[r, r + dr]$ is:

$$\rho_S(r,t)\, dr = |f(r,t)|^2\, 4\pi r^2\, dr$$

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